

Bulan 3: Django Fundamentals

Minggu 1: Django Setup & Basics

Hari 1: Django Installation & Project Setup

Topik: Django installation, startproject, project structure, settings.py

Proyek: MyFirstDjango

Setup proyek Django pertama

- Install Django dalam virtual environment
- Create project dengan django-admin startproject
- Explore project structure
- Run development server
- Understand settings.py basics

Hari 2: Django Apps & MVT Pattern

Topik: startapp, MVT (Model-View-Template), app structure

Proyek 1: HelloDjango

Aplikasi hello world Django

- Create app 'hello'
- Configure INSTALLED_APPS
- Create simple view
- Map URL ke view
- Return HttpResponse

Proyek 2: AboutPage

Aplikasi about page

- Create 'pages' app
- Multiple views (about, contact, home)
- URL routing untuk multiple pages
- Teknologi: Django apps, views, URLs

Hari 3: URL Routing & Views

Topik: path(), URL patterns, view functions, HttpResponseRedirect

Proyek: URLRouter

Sistem routing lengkap

- Create app 'blog'
- URL patterns dengan parameters
- path('post/int:id/')
- path('category/str:name/')
- View functions handle parameters
- Return dynamic responses

Hari 4: Templates Basics

Topik: Template directory, render(), template syntax, variables

Proyek: TemplateDemo

Demo template Django

- Configure TEMPLATES in settings
- Create templates folder
- render() function
- Pass context data ke template
- Display variables dalam template
- Template inheritance (base.html)

Hari 5: Template Tags & Filters

Topik: {% %}, {{ }}, for loop, if statement, filters

Proyek: ProductCatalog

Katalog produk dengan template

- Create Product list (hardcoded di view)
- Loop products dalam template
- If/else untuk stock availability
- Filters (upper, lower, date)
- Template tags (for, if)

- Teknologi: Django templates, template language

Hari 6: Static Files

Topik: STATIC_URL, STATIC_ROOT, {% static %}, collectstatic

Proyek: StyledBlog

Blog dengan static files

- Configure static files
- Create static/css, static/js, static/images
- Load static dalam template
- Add CSS file
- Add images
- Teknologi: static files management

Hari 7: Review & Mini Project

Mini Project: PersonalPortfolio

Website portfolio personal

- Multiple pages (home, about, projects, contact)
- Template inheritance (base.html)
- Static files (CSS, images, JS minimal)
- URL routing untuk semua pages
- Context data untuk projects
- Responsive layout (basic)
- Teknologi: Django apps, views, templates, static files, URL routing

Minggu 2: Django Models & Database

Hari 8: Models Basics

Topik: Model class, fields, CharField, IntegerField, DateField

Proyek: BookModel

Model untuk buku

- Create Book model (title, author, pages, published_date)
- makemigrations
- migrate

- Understand migrations
- Django admin setup

Hari 9: Model Fields & Options

Topik: TextField, BooleanField, DecimalField, field options (max_length, default, blank, null)

Proyek: ProductModel

Model produk e-commerce

- Product model (name, description, price, stock, is_available)
- Field validations
- default values
- blank=True, null=True
- **str** method
- Teknologi: Django models, field types

Hari 10: Django Admin

Topik: admin.site.register, ModelAdmin, list_display, search_fields

Proyek: AdminPanel

Customize Django admin

- Register models di admin
- list_display untuk columns
- list_filter
- search_fields
- ordering
- Custom ModelAdmin class
- Teknologi: Django admin customization

Hari 11: Model Relationships - ForeignKey

Topik: ForeignKey, on_delete, related_name, one-to-many

Proyek: BlogWithCategories

Blog dengan kategori

- Category model

- Post model dengan ForeignKey ke Category
- on_delete=CASCADE
- related_name='posts'
- Access posts dari category
- Teknologi: ForeignKey relationships

Hari 12: Model Relationships - ManyToMany

Topik: ManyToManyField, through model (optional)

Proyek: PostTagSystem

Sistem tag untuk posts

- Tag model
- Post model dengan ManyToManyField ke Tag
- Add/remove tags
- Query posts by tag
- Query tags by post
- Teknologi: ManyToMany relationships

Hari 13: QuerySets & ORM Basics

Topik: all(), filter(), get(), exclude(), order_by()

Proyek: ProductQuery

Query produk dengan ORM

- Model.objects.all()
- filter(price__lt=1000)
- exclude(stock=0)
- order_by('-created_at')
- get(id=1)
- Count, exists()
- Teknologi: Django ORM queries

Hari 14: Advanced Queries

Topik: Q objects, F expressions, aggregate, annotate

Proyek: AdvancedProductQuery

Query kompleks

- Complex queries dengan Q objects
- OR dan AND conditions
- F expressions untuk comparisons
- aggregate (Sum, Avg, Count)
- annotate untuk calculated fields
- Teknologi: Django ORM advanced queries

Minggu 3: Forms & User Input

Hari 15: Django Forms Basics

Topik: forms.Form, form fields, form validation

Proyek: ContactFormApp

Form kontak sederhana

- Create ContactForm dengan forms.Form
- CharField, EmailField
- Render form dalam template
- Handle GET dan POST
- is_valid()
- Teknologi: Django forms

Hari 16: Form Validation & Cleaning

Topik: clean_fieldname(), clean(), ValidationError

Proyek: RegistrationForm

Form registrasi dengan validasi

- Username, email, password fields
- Custom validation methods
- clean_email() untuk unique check
- clean() untuk password confirmation
- Display form errors
- Teknologi: Form validation, custom validators

Hari 17: ModelForms

Topik: forms.ModelForm, Meta class, fields, exclude

Proyek: ProductFormApp

CRUD produk dengan ModelForm

- ProductForm extends ModelForm
- Meta class dengan model & fields
- Create product dari form
- Auto validation dari model
- Display form dengan `{{ form.as_p }}`
- Teknologi: ModelForm

Hari 18: Form Widgets & Customization

Topik: widgets, Textarea, Select, CheckboxInput, widget attrs

Proyek: StyledProductForm

Form dengan custom widgets

- Custom widget attrs (class, placeholder)
- Textarea untuk description
- DateInput dengan type='date'
- Custom CSS classes
- Bootstrap form styling
- Teknologi: Form widgets customization

Hari 19: Handling File Uploads

Topik: FileField, ImageField, request.FILES, MEDIA_URL, MEDIA_ROOT

Proyek: ImageUploader

Upload gambar produk

- Add ImageField ke Product model
- Configure MEDIA settings
- Handle request.FILES
- Save uploaded file
- Display uploaded images

- File validation
- Teknologi: File uploads, media files

Hari 20: Form Processing & CRUD Operations

Topik: Create, Read, Update, Delete dengan forms

Proyek: TaskCRUD

Full CRUD untuk tasks

- Create task dengan form
- List all tasks
- Update task (pre-fill form)
- Delete task (confirmation)
- Redirect after success
- Messages framework
- Teknologi: CRUD operations, forms, messages

Hari 21: Review & Mini Project

Mini Project: BlogPlatform

Platform blog lengkap dengan CRUD

Models:

- Category (name, slug)
- Post (title, content, category, author, created_at, updated_at, published)
- Comment (post, author_name, content, created_at)

Features:

- List all posts dengan pagination
- Post detail dengan comments
- Create/Edit/Delete post (ModelForm)
- Filter posts by category
- Add comment to post
- Search posts
- Django admin untuk manage
- Static files untuk styling

- Teknologi: Models, Forms, Templates, CRUD, QuerySets, Admin

Minggu 4: Authentication & Authorization

Hari 22: User Authentication Setup

Topik: Django User model, django.contrib.auth

Proyek: AuthSetup

Setup authentication sistem

- Explore built-in User model
- Create superuser
- Login admin panel
- Understand authentication backend
- User model fields

Hari 23: User Registration

Topik: UserCreationForm, create user, password hashing

Proyek: UserRegistration

Sistem registrasi user

- Import UserCreationForm
- Create registration view & template
- Handle form submission
- Create new user
- Auto login after registration
- Redirect to home
- Teknologi: UserCreationForm, authentication

Hari 24: Login & Logout

Topik: LoginView, LogoutView, login_required decorator

Proyek: AuthSystem

Login/Logout sistem

- LoginView dengan template
- LogoutView

- @login_required decorator
- Protect views
- LOGIN_URL setting
- Next parameter
- Teknologi: Login/Logout views, decorators

Hari 25: User Profile & Custom User

Topik: One-to-One with User, extending User model

Proyek: UserProfile

Profile untuk user

- Profile model dengan OneToOneField
- Additional fields (bio, avatar, phone)
- Signal untuk auto create profile
- Update profile view
- Display profile
- Teknologi: OneToOne relationship, signals

Hari 26: Permissions & Groups

Topik: Permissions, Groups, user.has_perm(), permission_required

Proyek: PermissionSystem

Sistem permission

- Create custom permissions
- Create groups (Admin, Editor, Viewer)
- Assign permissions to groups
- Add users to groups
- Check permissions dalam view
- @permission_required decorator
- Teknologi: Permissions, Groups

Hari 27: Password Management

Topik: PasswordChangeView, PasswordResetView, email backend

Proyek: PasswordManagement

Ubah & reset password

- Password change view
- Password change form
- Password reset (email simulation)
- Password reset confirm
- Success messages
- Teknologi: Password views

Hari 28: Review & Final Project Bulan 3

Final Project: TaskManagementWeb

Web app task management dengan auth

Models:

- User (built-in)
- UserProfile (bio, avatar)
- Project (name, description, owner, members)
- Task (title, description, project, assigned_to, status, priority, due_date)

Features:

- User registration & login
 - User profile management
 - Create/Edit/Delete projects (owner only)
 - Add members to projects
 - Create/Edit/Delete tasks
 - Assign tasks to members
 - Filter tasks (by status, priority, assigned user)
 - Dashboard dengan statistics
 - Permission checks (only members can view)
 - Django admin panel
 - Responsive templates
 - Teknologi: Auth, Permissions, Models, Forms, CRUD, QuerySets, Signals
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Progress Check Bulan 3: ☒ Django project & app structure ☒ URL routing & views ☒ Templates & template language ☒ Static files management ☒ Models & migrations ☒ Django ORM & QuerySets ☒ Forms & ModelForms ☒ File uploads ☒ CRUD operations ☒ Authentication & Authorization ☒ Permissions & Groups

Total Proyek: 55+ proyek & 3 major projects **Siap lanjut:** Bulan 4 - Django Advanced Features